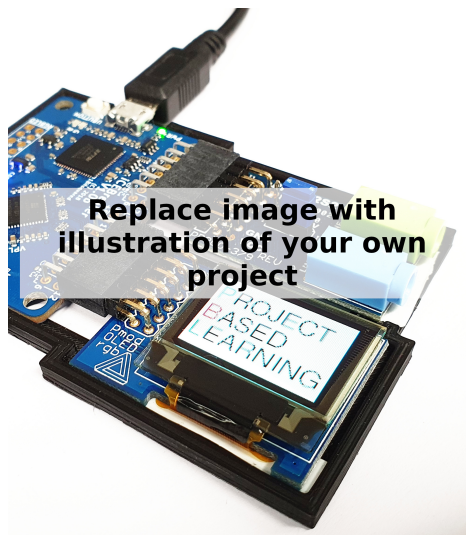


DEPARTMENT OF INFORMATION TECHNOLOGY AND
ELECTRICAL ENGINEERING

Spring Semester 2022

L^AT_EX Report Template

Bachelor Thesis / Semester Thesis / Master Thesis

Your Name
your@name.com

March 2022

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Acknowledgements

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Abstract

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The **abstract** is the presentation card of your work, it is fundamental to describe your contribution and relevant results in few lines of text. It must fit in one page and should include:

- Short description of your topic or use case;
- Short description of existing technology, work, product and the scientific research challenges still unsolved;
- Description of your work and your contribution;
- A short summary of the most relevant results of your work and the conclusions you draw from it.

Declaration of Originality

I hereby confirm that I am the sole author of the written work here enclosed and that I have compiled it in my own words. Parts excepted are corrections of form and content by the supervisor. For a detailed version of the declaration of originality, please refer to Appendix B

Your Name,
Zurich, March 2022

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List of Acronyms

AES	Advanced Encryption Standard
ASIC	Application-Specific Integrated Circuit
DES	Data Encryption Standard
DVI	Device Independent File Format
ECC	Elliptic Curve Cryptography
ECDSA	Elliptic Curve Digital Signature Algorithm
EPS	Encapsulated PostScript
FPGA	Field Programmable Gate Array
IC	Integrated Circuit
IIS	Integrated Systems Laboratory
LED	Light-Emitting Diode
NIST	National Institute of Standards and Technology
PBL	Center for Project-Based Learning
PDF	Portable Document Format
WYSIWYG	What You See Is What You Get

Chapter 1

Introduction

Give an overview of the problem, and put your work into a bigger context. Motivate the questions addressed in this work and summarize your contributions.

If you want to *highlight* a word or term use `\emph{}`.

If you work with many acronyms we recommend the use of an acronym library. You can define acronyms in the lower part of the glossaries.tex file. When first using the acronym it will be introduced Center for Project-Based Learning (PBL), after that the shortform will be displayed PBL.

1.1. Motivation

Give a broader context on why your thesis is important, what problem is solved by it, why is it needed,?

1.2. Objective

what was the objective of your thesis? also add your research questions here

1.2.1. Research Questions

1. first research qestion...
2. second reserach question...

1. Introduction

1.2.2. Outline

At the end of this chapter, give an outline and describe the organization of your thesis

Chapter 2

Related Work

Describe relevant aspects of related work, make sure to have a good covering of your approach/design with others who did similar things. You will need to compare this related work with your results in the Discussion chapter. We expect a lot of citations [1] here.

Make sure to also give the results of the related work relevant to your thesis presented here, including numbers. Often, at the end of this section an overview table makes sense, where you compare related work for different aspects related to your own thesis.

Chapter 3

Theory / Background

Explain theoretical concepts and background you used in your work, and give some background to understand the significance of your work in a larger context. Find a bit of a trade-off between creating a textbook and giving all necessary information to understand your thesis.

This chapter also includes the description of existing work that you used, served, as a basic platform to develop your project. For example, if you programmed a specific MCU, here you can report a short description of the MCU-family and its main features.

If you introduce equations in your theory chapter (or any other chapter) use the following syntax to get numbered equations that you can refer to as equation 3.1:

$$y_i = mx_i + b \tag{3.1}$$

Explain all the variable names that you used in your equation. In this case x_i are the inputs to a linear equation in slope-intercept form with slope m and intercept b where the results are called y_i with $i \in [0, 5]$.

Chapter 4

Hardware / Firmware / Algorithm Implementation

Describe how you implemented the functionality, be that hardware, firmware, algorithms etc. Always important: also describe why the design choices are as they are! Also feel free to rename the chapter so something more telling, e.g. *Implementation of Gesture Recognition on uBean*.

Chapter 5

Results

First describe your measurement setup and settings (so it could be reproduced) and afterwards describe the results you got from your experiments. Putting results in perspective and interpretation do not belong in this chapter (put that into the discussion)

5.1. Method

experimental setup, used hardware and software, etc. Always rename this section appropriately, e.g. *Range measurement setup*.

5.2. Results with Method 1

Give the results obtained with the previously described method, rename when usefull, e.g. *Range measurement results*.

Chapter 6

Discussion

Draw your conclusions from the results you achieved and summarize your contributions. Comparisons (e.g., of hardware figures) with related work are also appropriate here. Point out things that could or need to be investigated further.

Chapter 7

Conclusion and Future Work

Short chapter on the conclusions you can draw from your work, and what is still to be done (and what contributions the students after you could make).

Appendix A

Task Description

Include the task description **pdf** you got from your assistant(s) with the `\includepdf` command.

Appendix B

Declaration of Originality

Include the declaration of authorship with the `\includepdf` command (sign it and scan it). For more information about plagiarism, please visit <https://www.ethz.ch/students/en/studies/performance-assessments/plagiarism.html>

- **English version:** <https://www.ethz.ch/content/dam/ethz/main/education/rechtliches-abschluesse/leistungskontrollen/declaration-originality.pdf>
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Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Declaration of originality

The signed declaration of originality is a component of every semester paper, Bachelor's thesis, Master's thesis and any other degree paper undertaken during the course of studies, including the respective electronic versions.

Lecturers may also require a declaration of originality for other written papers compiled for their courses.

I hereby confirm that I am the sole author of the written work here enclosed and that I have compiled it in my own words. Parts excepted are corrections of form and content by the supervisor.

Title of work (in block letters):

This is a sample title

Authored by (in block letters):

For papers written by groups the names of all authors are required.

Name(s):

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With my signature I confirm that

- I have committed none of the forms of plagiarism described in the 'Citation etiquette' information sheet.
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I am aware that the work may be screened electronically for plagiarism.

Place, date

Zurich, 01.01.2000

Signature(s)

First Student Signature
Second Student Signature

For papers written by groups the names of all authors are required. Their signatures collectively guarantee the entire content of the written paper.

Appendix C

(optional) File Structure

Describe how the project directories/files are organized, e.g.:

```
/
├── README ..... A README with some general information about the project.
├── 01_report ..... The source files of the project report.
├── 02_presentation ..... The source files of the presentation.
└── 03_designflow ..... Some designflow-specific files.
```

What needs to be done to run an RTL simulation (stimuli generation, compilation...)?

Appendix D

(optional) Datasets

If you have a data set comprising several test images, you could depict and describe them here. Use a simple naming scheme such that you can easily refer to certain elements of this data set in the text.

Appendix E

(optional) More Evaluation Results

If you conducted an extensive evaluation you could move surplus graphs/results to the appendix.

Glossary

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